RIYADH HEATREADY

Urban growth and surface heat screening

A satellite-based priority map for southeast Riyadh

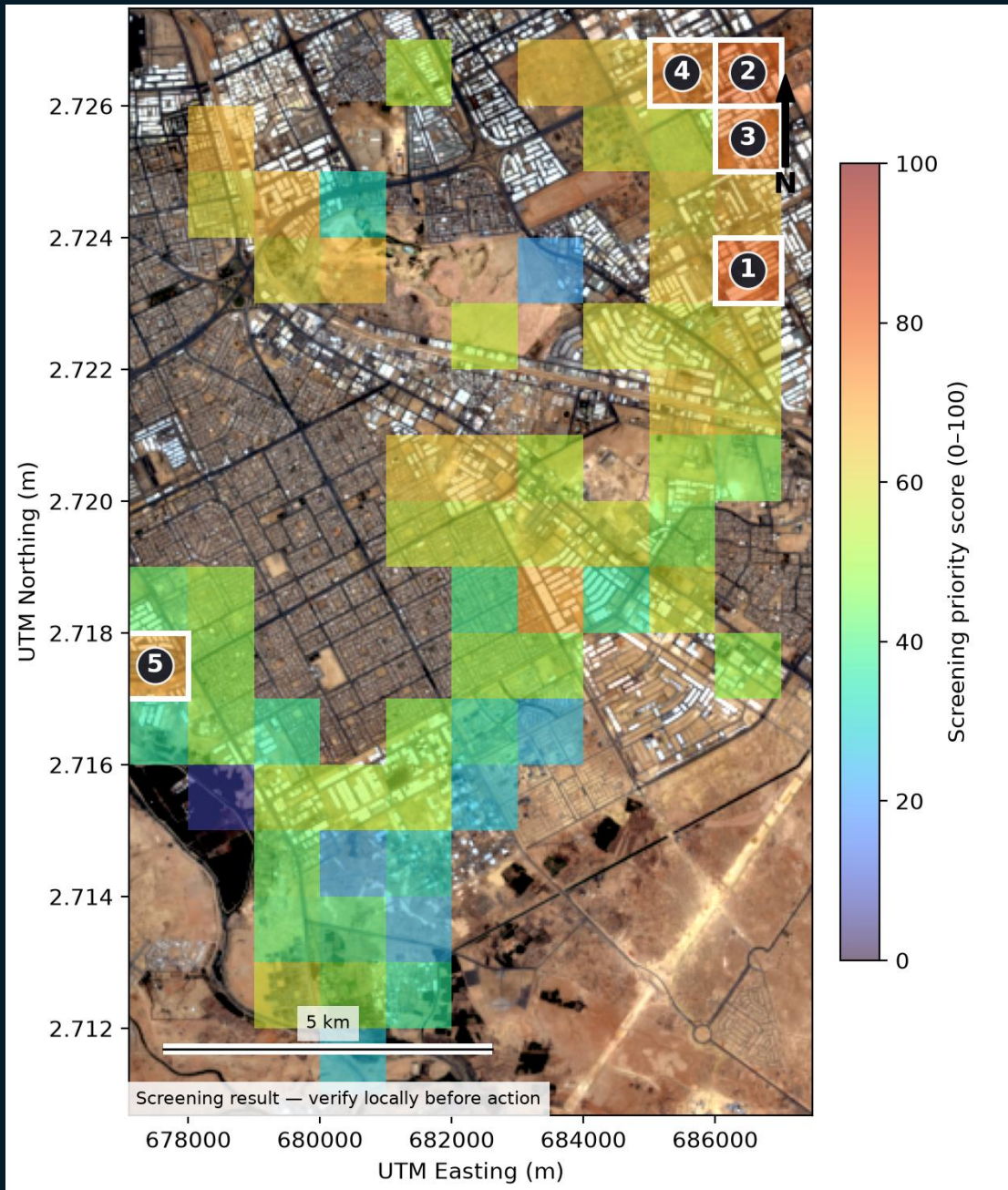
THE QUESTION

Where should a municipal planning team inspect first?

Team Milkaholics

Yusuf Karanib and Raed Mardoum

PRIORITY MAP



Team roles

Yusuf Karanib

Project lead

- Leads the project direction and final submission.
- Presents the data, analysis, score and demonstration.
- Focuses on the data workflow and reproducibility.

Raed Mardoum

Validation and planning workflow

- Presents the validation results and limitations.
- Explains how planners should use the ranking.
- Presents the expected impact and pilot plan.

A faster way to choose areas for inspection

168.30 km²
SOUTHEASTERN RIYADH STUDY AREA

1 km
TRANSPARENT SCREENING CELLS

1 user
MUNICIPAL PLANNING + HEAT TEAM

1 action
INSPECT LOCALLY BEFORE INTERVENTION

PROBLEM

Growth, heat, population and vegetation sit in separate layers. A planner still needs a short, explainable list of places to inspect.

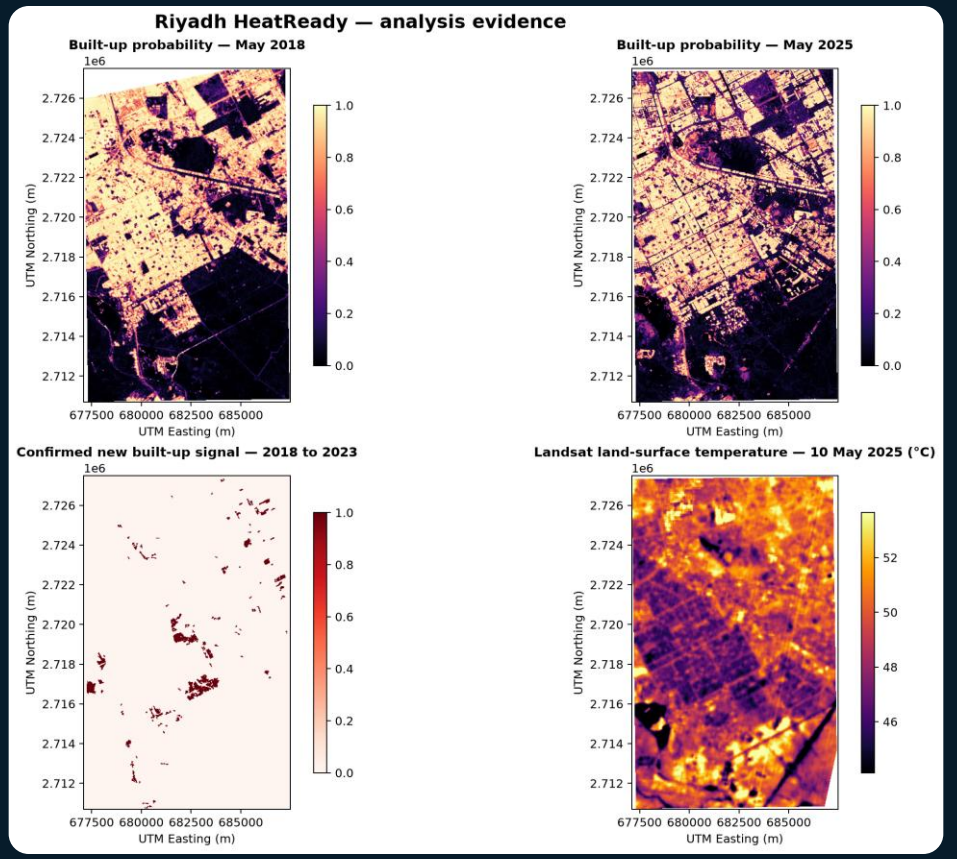
OUTPUT

HeatReady combines the layers into a ranked list of 1 km cells for local review. It does not choose an intervention.

Current limitation: we have not interviewed municipal users. A pilot must test whether this fits their workflow.

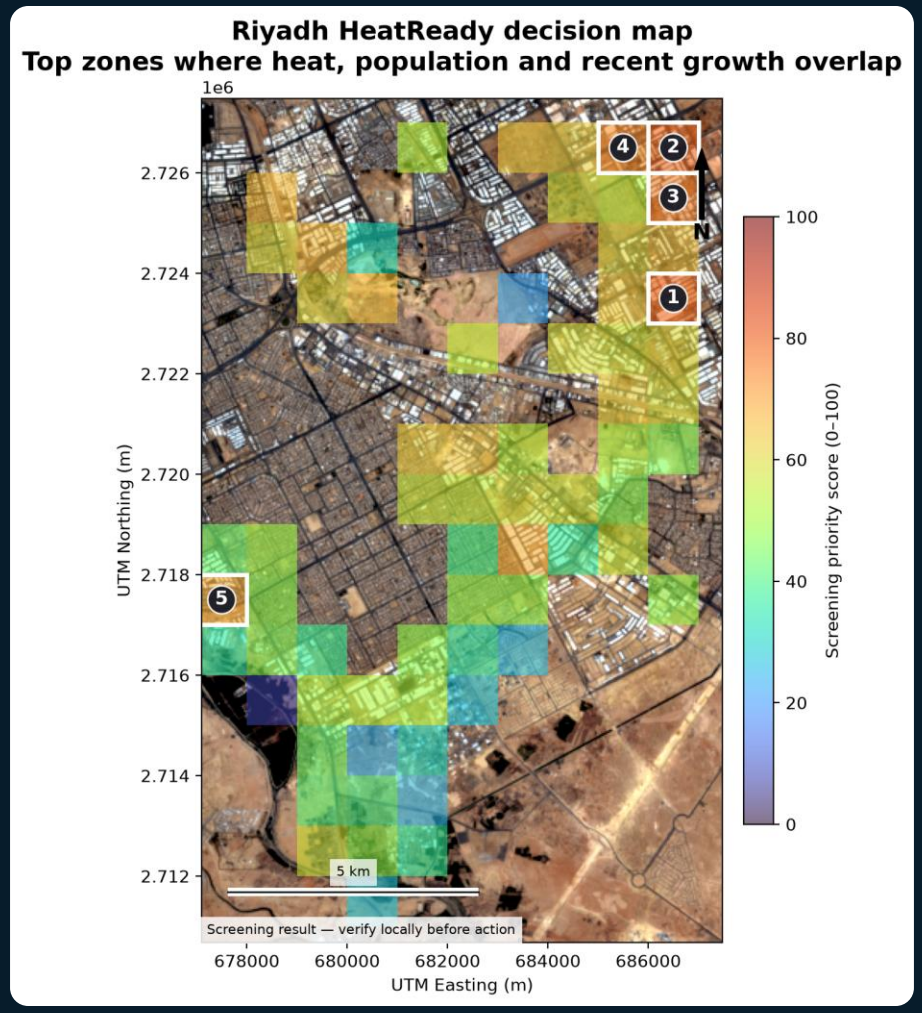
Data used in the analysis

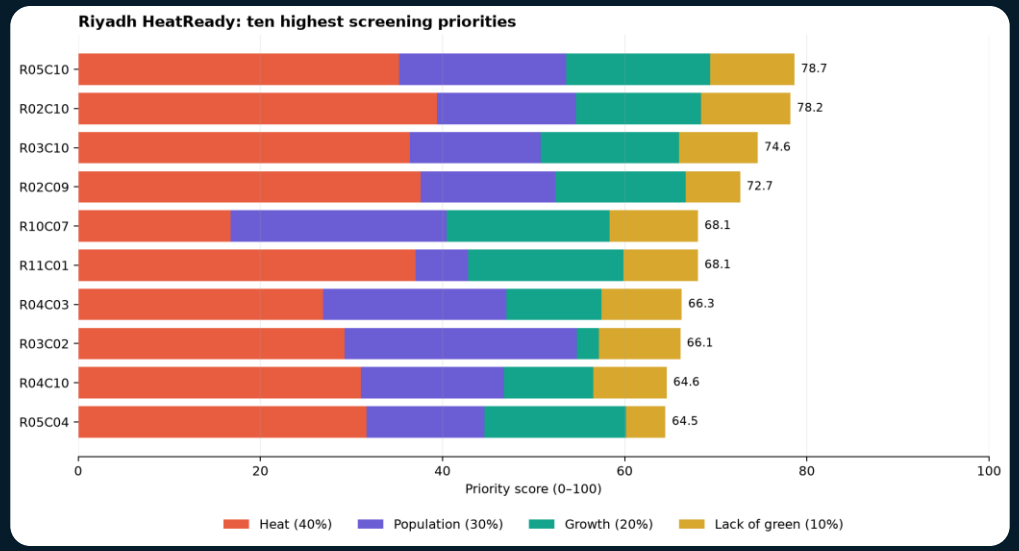
Sentinel-2	Optical reflectance, vegetation and built features
Sentinel-1	Radar structure to reduce bright-desert confusion
Annual LULC	Six maps propose 2018–2023 built-up change
Landsat 8	Quality-masked 2025 land-surface temperature
WorldPop	Modelled 2025 population exposure
Planet Tanager	426-band material context and failure-mode test



1. Annual maps flag possible growth. 2. Sentinel-1 and Sentinel-2 check the built-up evidence. 3. Heat, population and vegetation produce the ranking.

Priority map and supporting files





3.74 km²
CONFIRMED GROWTH SIGNAL

14 / 67
HIGH / ELIGIBLE CELLS

78.7
TOP SCORE: R05C10

The HTML report shows the map and top-five table. CSV, GeoJSON and GeoTIFF files are included separately.

Validation results and main limitations

0.88

F1 AGREEMENT

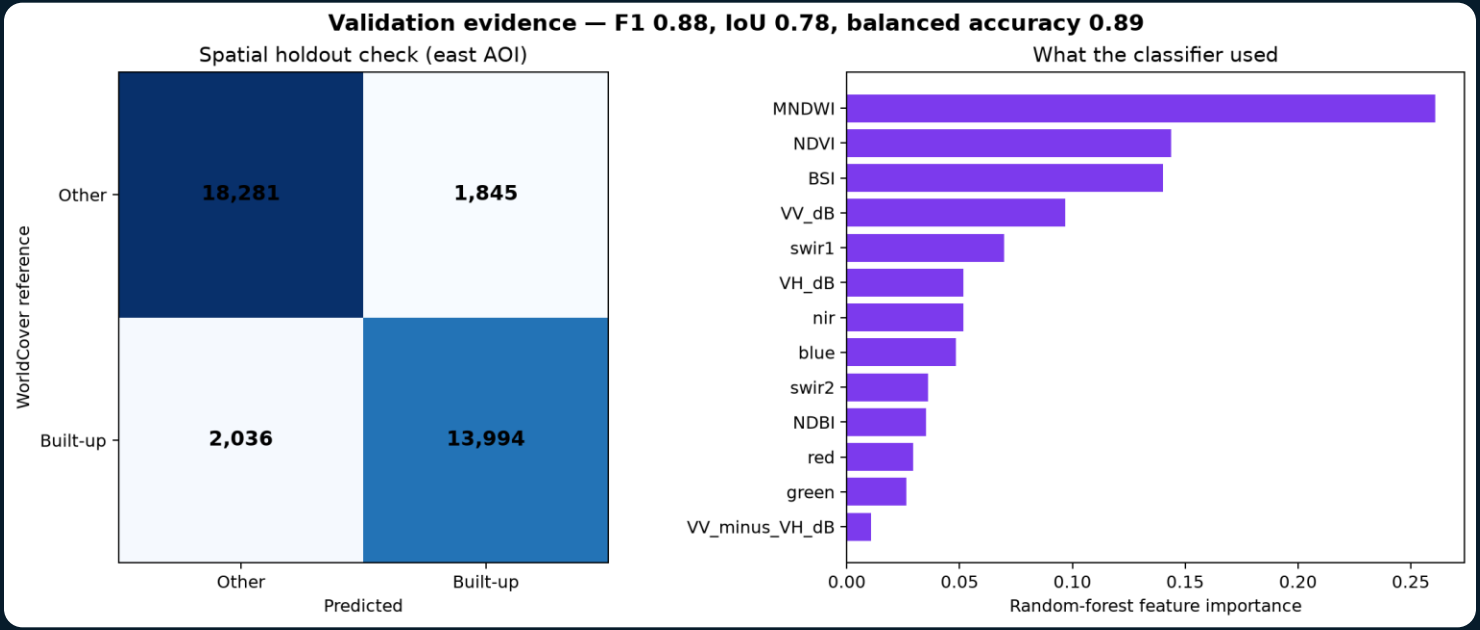
0.78

INTERSECTION OVER UNION

0.89

BALANCED ACCURACY

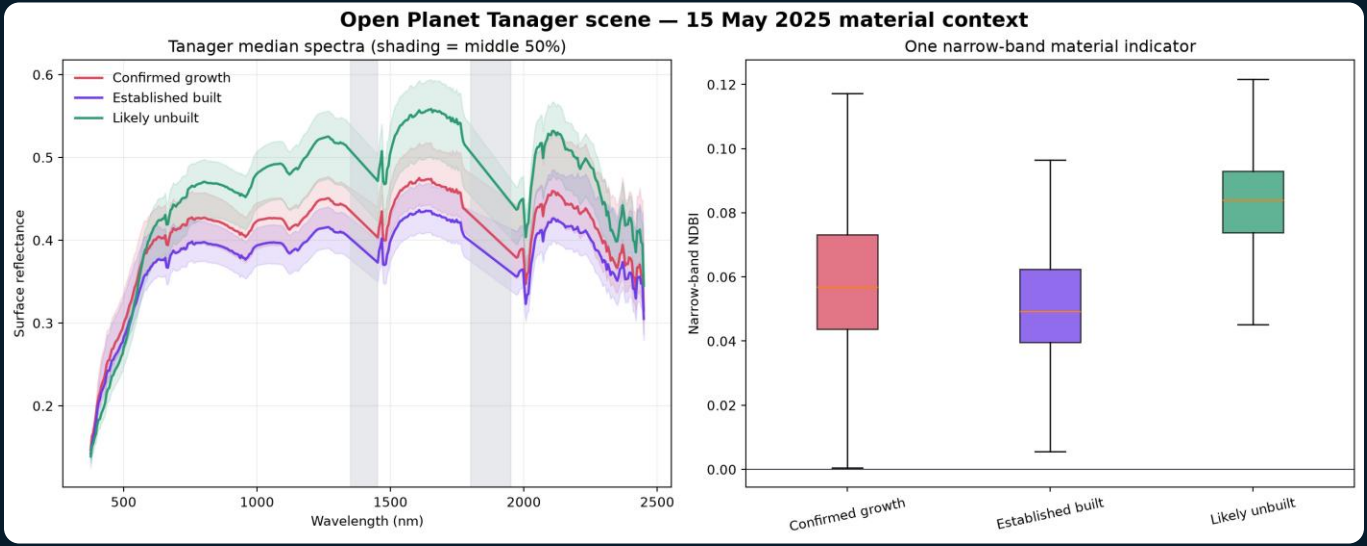
These compare the model with a spatially separate WorldCover reference. They are not field accuracy.



MAIN LIMITATIONS

- Land-surface temperature is not air temperature or personal heat exposure.
- WorldPop is modelled; it is not a live census or worker count.
- Bright desert and roofs can still confuse classification.
- Score weights are PoC assumptions that a real user must review.

How HeatReady differs from a single-index map



METHOD

- Annual land-cover maps flag possible growth.
- Optical and radar evidence check whether it still looks built-up.
- Each priority score keeps its visible components and GIS evidence.

Limit: one Tanager scene adds spectral context. It cannot prove change or identify a material.

POSSIBLE BENEFIT

A planner can review one ranked list instead of comparing several separate layers. This still needs user testing.

Proposed municipal pilot

1

Interview

Confirm the decision, current workflow and acceptable false alarms with at least three real users.

2

Field check

Inspect the top five zones, record useful results and false alarms, and compare with local plans.

3

Adjust

Agree on weights, add worker exposure and city-plan constraints, then rerun the ranking.

4

Repeat

Test the same process in another Riyadh area before making an operational claim.

PILOT REQUEST

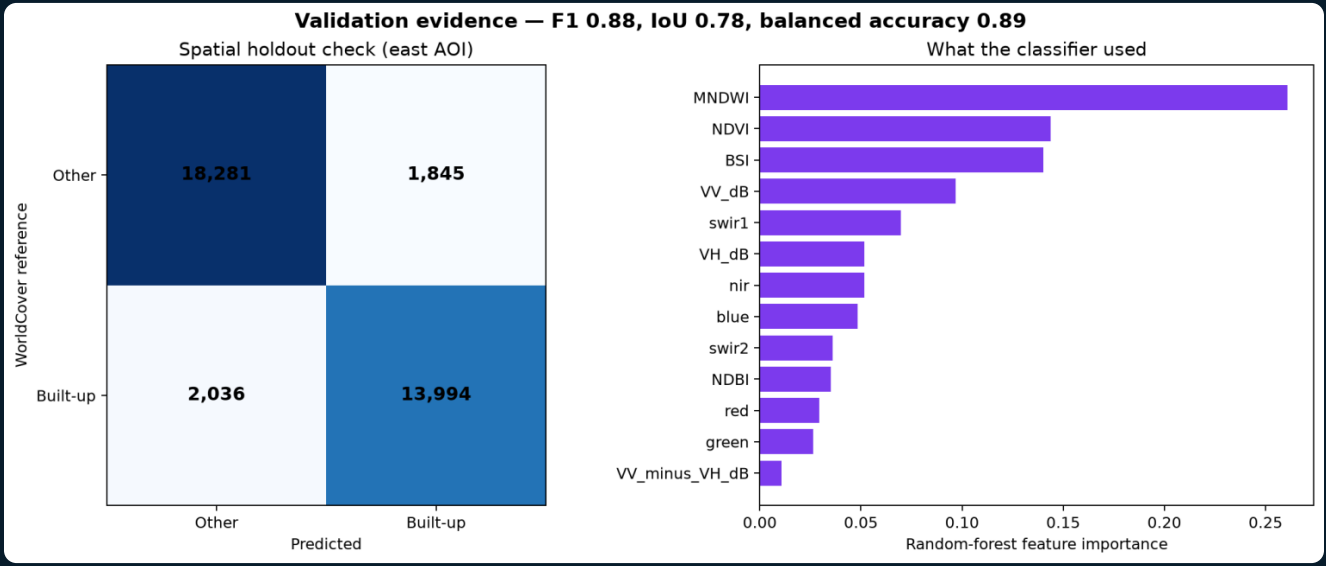
Test HeatReady with one municipal team and measure useful findings, false alarms and time spent.

Data sources and dates

Source	Dates used	Role	Access / licence
Sentinel-2 L2A	11 May 2018; 10 May 2021; 9 May 2025	Optical features, masks, vegetation	Copernicus open access
Sentinel-1 RTC	10 May 2018; 6 May 2021; 9 May 2025	Radar structure	Microsoft PC; CC BY 4.0
IO/Esri annual LULC	2018–2023	Trend + candidate endpoint change	CC BY 4.0
Landsat 8 C2 L2	10 May 2025	Land-surface temperature	US public domain
ESA WorldCover	2021	Spatial holdout reference	CC BY 4.0
WorldPop Global2	2025	Modelled population	CC BY 4.0
Planet Tanager	15 May 2025	Spectral context + sensitivity test	Open urban archive; CC BY 4.0

Item IDs, processing steps and licences are documented in the repository. Satellite 813 imagery was not used.

Validation evidence and limits



Training area: west Tuning area: middle Test area: east

WHAT THE TEST SHOWS

- The east-side holdout was compared with WorldCover.
- F1 0.88, IoU 0.78 and balanced accuracy 0.89.
- The test contains 36,156 reference pixels.

WHAT IT DOES NOT SHOW

- Field accuracy.
- Air temperature or health outcomes.
- Municipal approval or user acceptance.

Next: field and user review of the top-ranked cells.